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Specification of

RISC-V 64-Bit Single-Core Microcontroller

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Introduction

Table 1.1: History / Revision – Chapter 1

Version	Date	Author	Remarks
1.0	January 22, 2026	Y. Sangale	Initial template draft
1.0	April 14, 2026	Y. Sangale	FPU Specs draft

1.1 Purpose of Document

This document defines the first specification of the Reduced Instruction Set Computer - Five (RISC-V) single-instruction-bit microcontroller, covering architectural, electrical, and functional aspects. Revision 1.1 adds the Floating-Point Unit (FPU) specification as an additional design element.

1.2 Scope

The specification outlines the design decisions, Instruction Set Architecture (ISA) subset, interfaces, and architectural details of the RISC-V System-on-Chip. This revision extends the scope to include the IEEE 754-2008 compliant double-precision FPU.

1.3 Reference Documents

- RISC-V Unprivileged ISA Specification, Volume I
- RISC-V Privileged Architecture Specification
- IEEE 754-2008: Standard for Floating-Point Arithmetic
- <https://docs.riscv.org/reference/isa/unpriv/f-st-ext.html>
- RWU “Specification Document Preparation Guidelines”